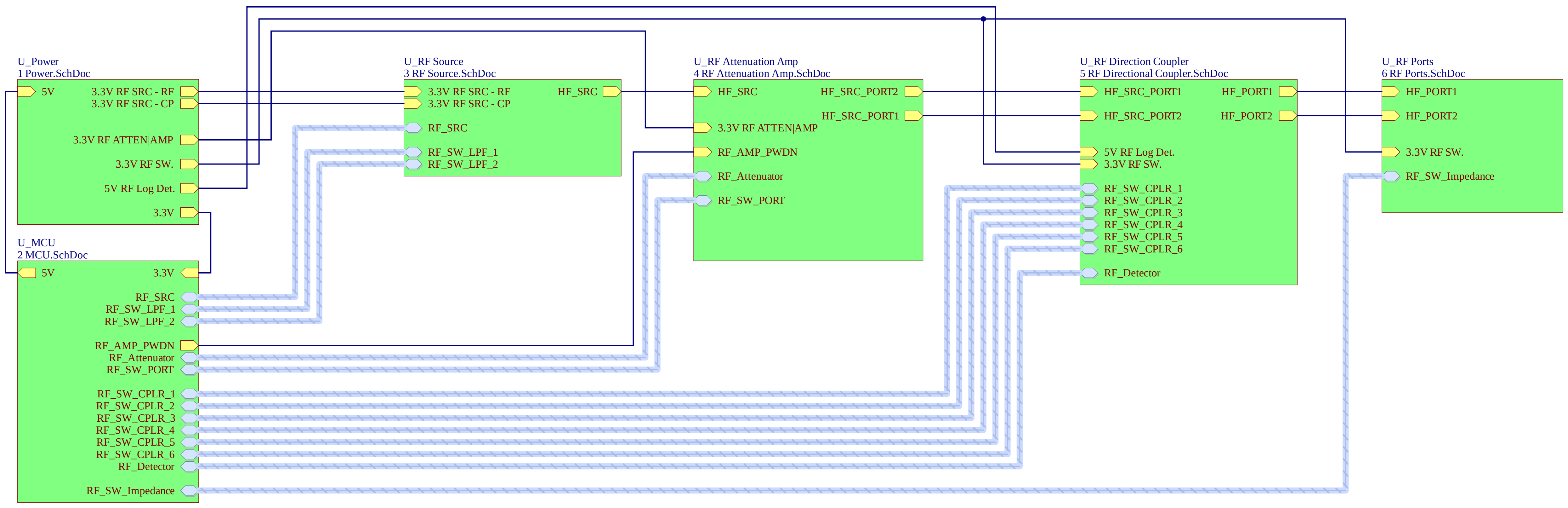
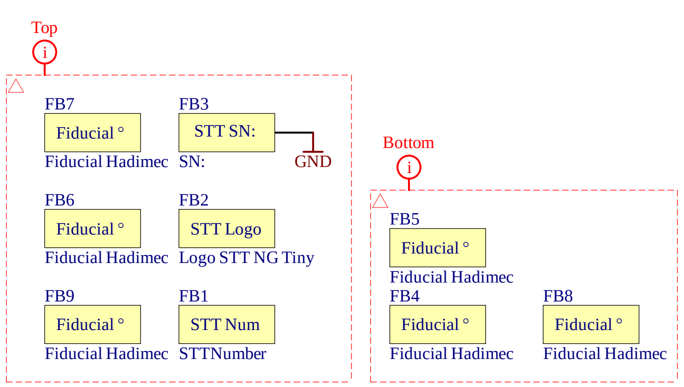
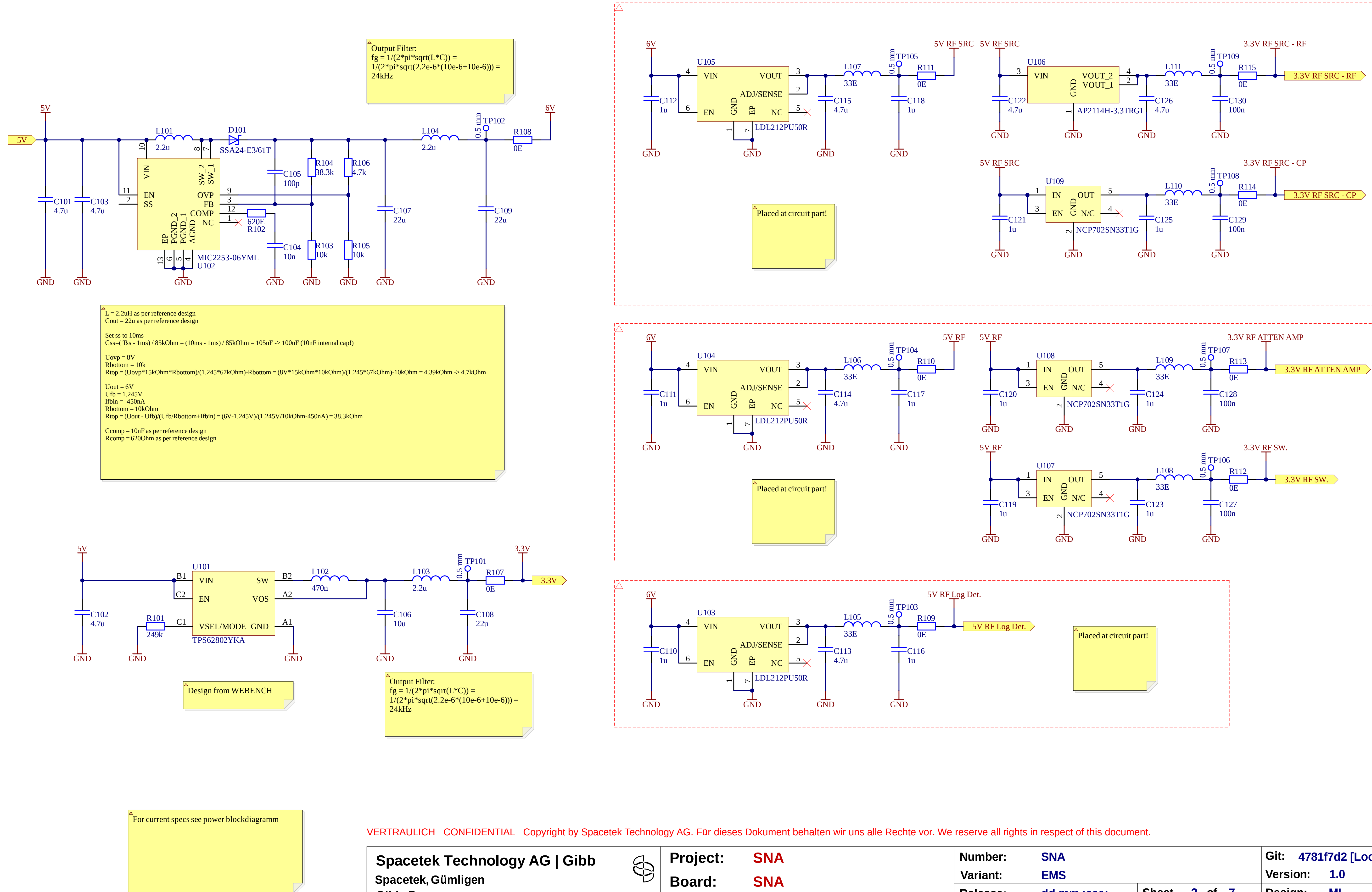




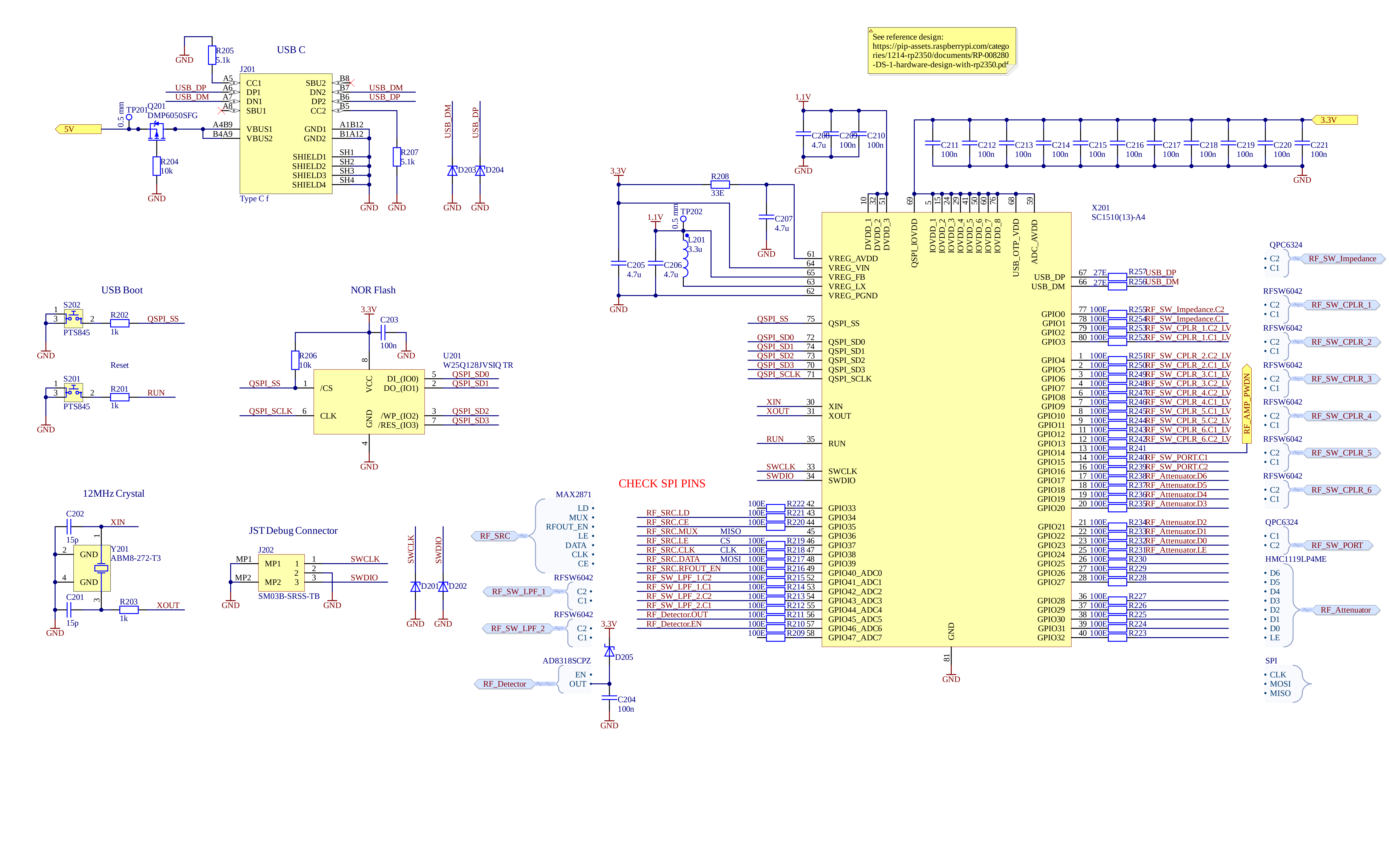
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** Overview	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy **Sheet** 1 **of** 7 **Size:** A3	**Git:** 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

File: https://github.com/ml3rc/SNA.git — 0 Overview.SchDoc



Spacetek Technology AG | Gibb **Spacetek, Gümligen** **Gibb, Bern**	**Project:** SNA **Board:** SNA **Page:** Power	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Git:** 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML
		Sheet 2 of 7 **Size:** A3	

File: https://github.com/ml3rc/SNA.git 1 Power.SchDoc



Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

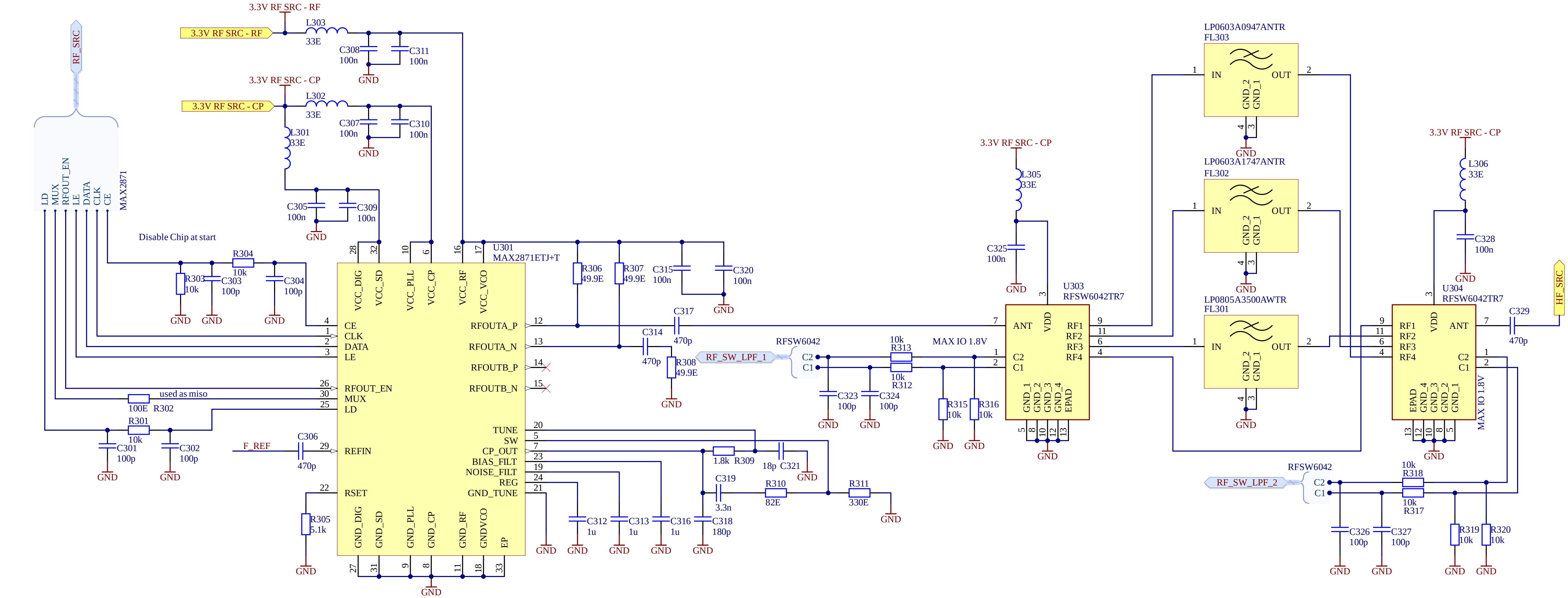
Project:	SNA	Number:	SNA
Board:	SNA	Variant:	EMS
Page:	MCU	Release:	dd.mm.yyyy
		Sheet	3 of 7
		Size:	A3

Git:	4781f7d2 [Locally M
Version:	1.0
Design:	ML
Check:	ML

File: https://github.com/ml3rc/SNA.git

2 MCU.SchDoc

See LibreVNA Schematics for reference



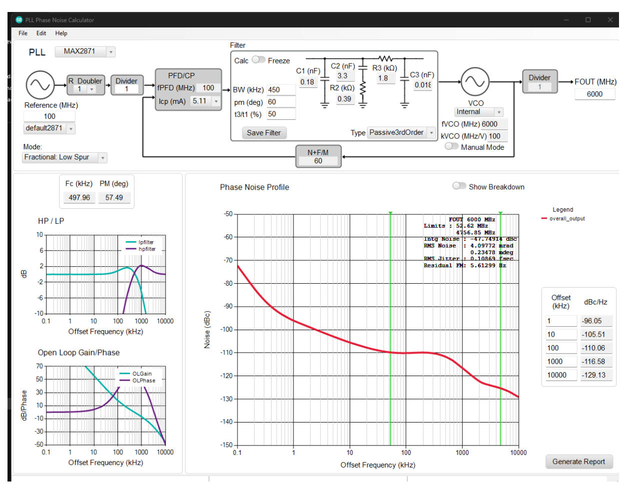
Rset = 5.1kOhm as per datasheet

C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

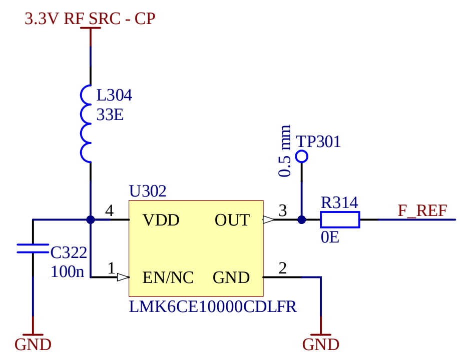
The loop filter can be calculated with:
https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

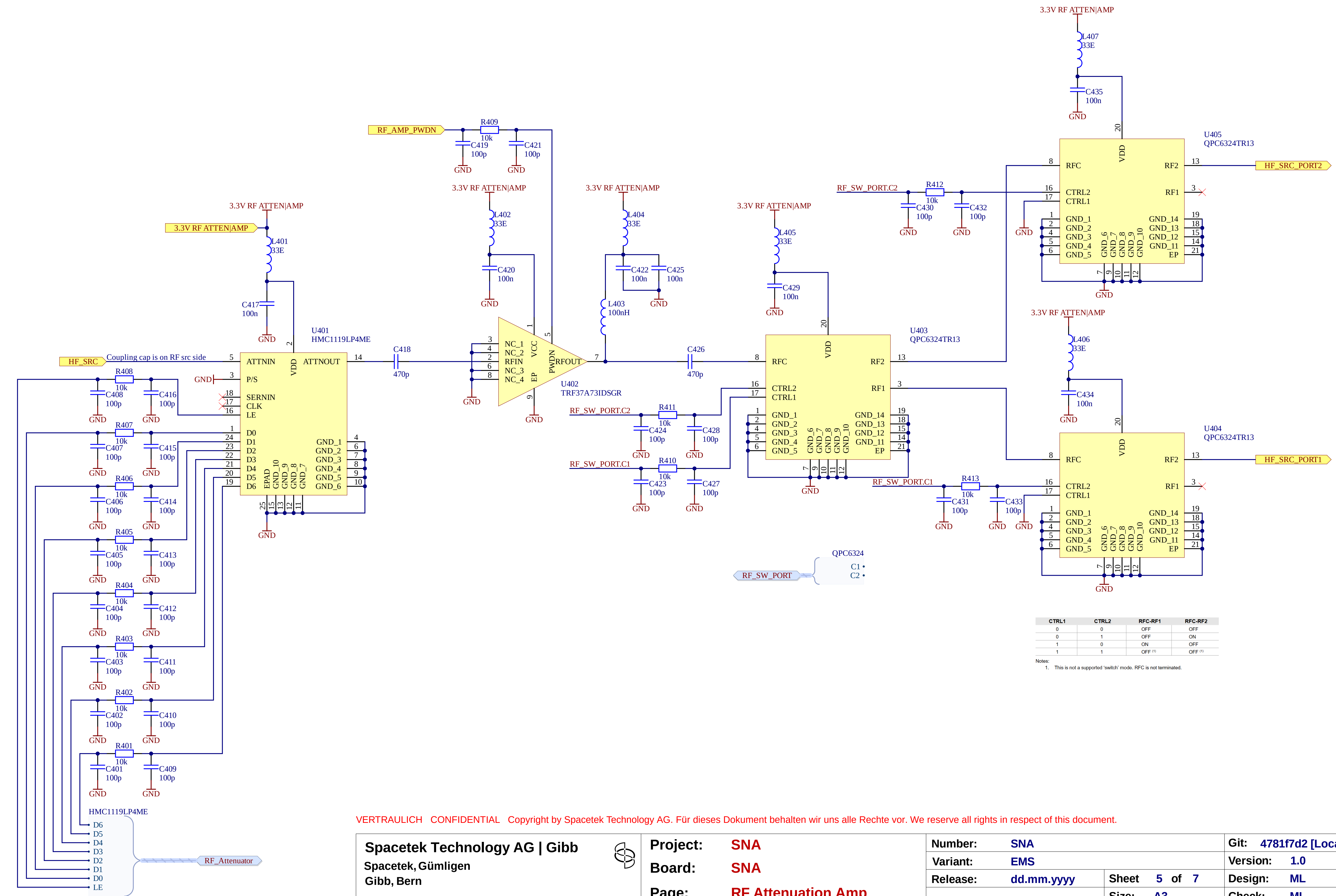
R_Crf_coupling = ((1)/(2*π*50MHz*470pF)) = 6.8Ohm



100MHz REF Clk.



Spacetek Technology AG | Gibb	**Project:** SNA	**Number:** SNA	**Git:** 4781f7d2 [Locally M
Spacetek, Gümligen	**Board:** SNA	**Variant:** EMS	**Version:** 1.0
Gibb, Bern	**Page:** RF Source	**Release:** dd.mm.yyyy — **Sheet** 4 **of** 7	**Design:** ML
		Size: A3	**Check:** ML
	File: https://github.com/ml3rc/SNA.git		3 RF Source.SchDoc



CTRL1	CTRL2	RFC-RF1	RFC-RF2
0	0	OFF	OFF
0	1	OFF	ON
1	0	ON	OFF
1	1	OFF [1]	OFF [1]

Notes:

1. This is not a supported 'switch' mode. RFC is not terminated.

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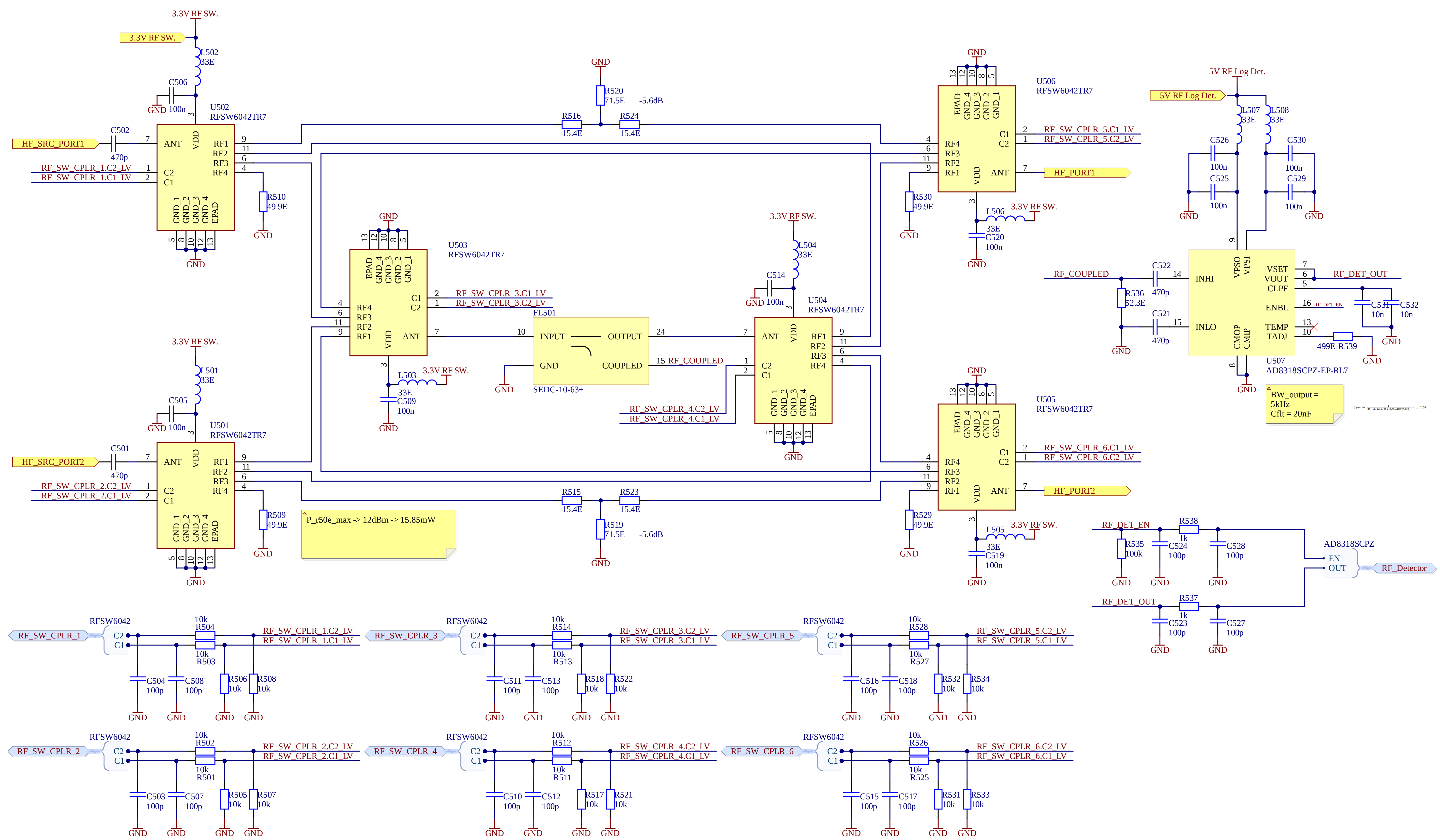
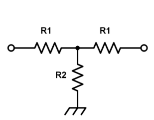
Gibb, Bern

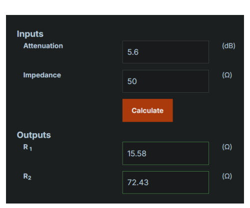
Project:	SNA	Number:	SNA	Git:	4781f7d2 [Locally M
Board:	SNA	Variant:	EMS	Version:	1.0
Page:	RF Attenuation Amp	Release:	dd.mm.yyyy Sheet 5 of 7	Design:	ML
			Size: A3	Check:	ML

File: https://github.com/ml3rc/SNA.git 4 RF Attenuation Amp.SchDoc



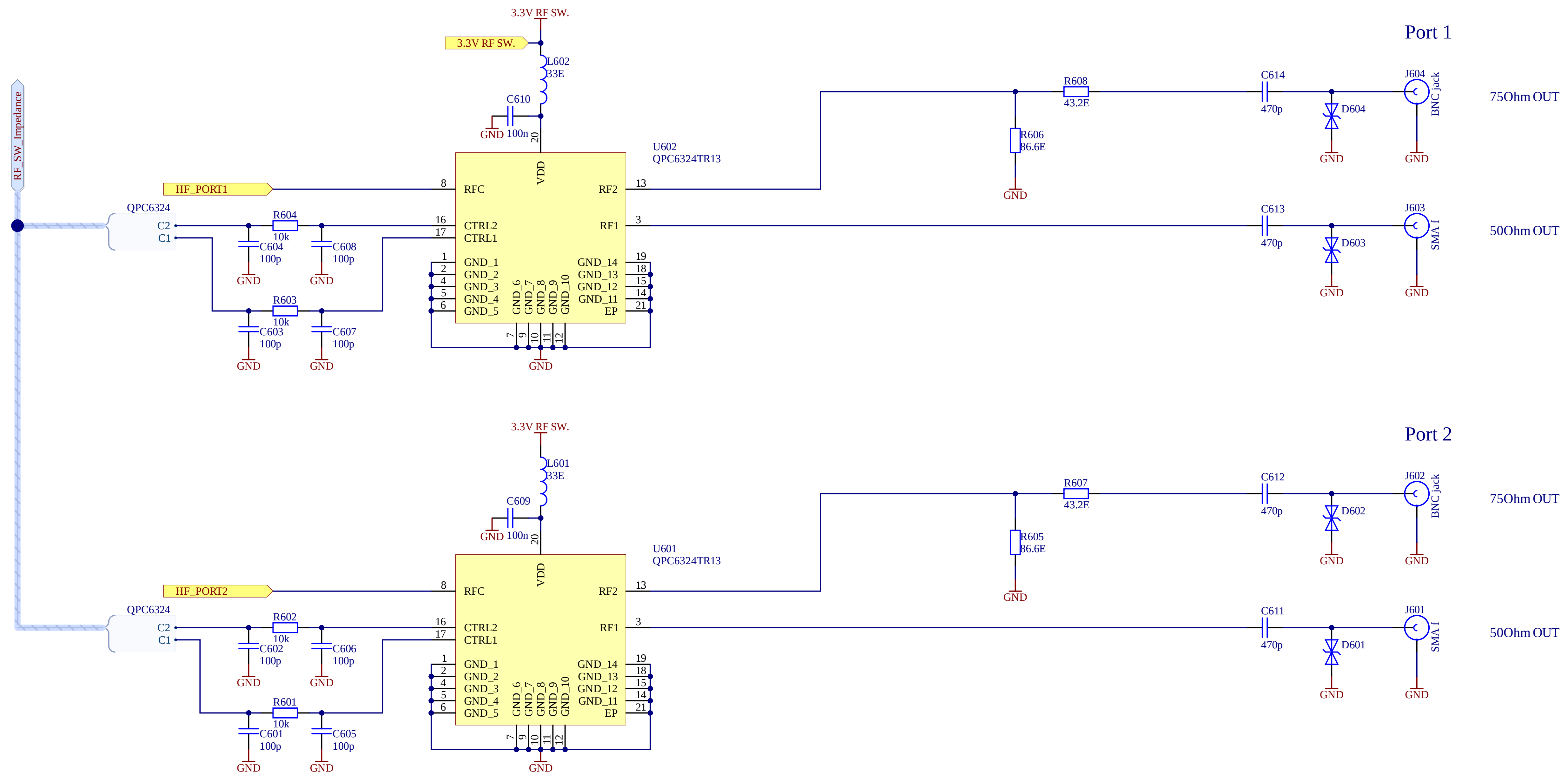
T-Pad calculated with:
https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/





Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** RF Directional Coupler	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy **Sheet** 6 **of** 7 **Size:** A3	**Git:** 4781f7d2 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

File: https://github.com/ml3rc/SNA.git — 5 RF Directional Coupler.SchDoc



50-75 Matching Pad:
solve({75=((1)/(((1)/(Rp))+((1)/(50))))+Rs,50=((1)/(((1)/(75+Rs))+((1)/(Rp))))})
Rp = 86.6Ohm
Rs = 43.3Ohm
Insertion Loss = 5.72dB

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Spacetek, Gümligen
Gibb, Bern

Project: SNA	**Number:** SNA		**Git:** 4781f7d2 [Locally M
Board: SNA	**Variant:** EMS		**Version:** 1.0
Page: RF Ports	**Release:** dd.mm.yyyy	**Sheet** 7 **of** 7	**Design:** ML
		Size: A3	**Check:** ML

File: https://github.com/ml3rc/SNA.git 6 RF Ports.SchDoc